

# Computer Networks

## *[Network Topology]*

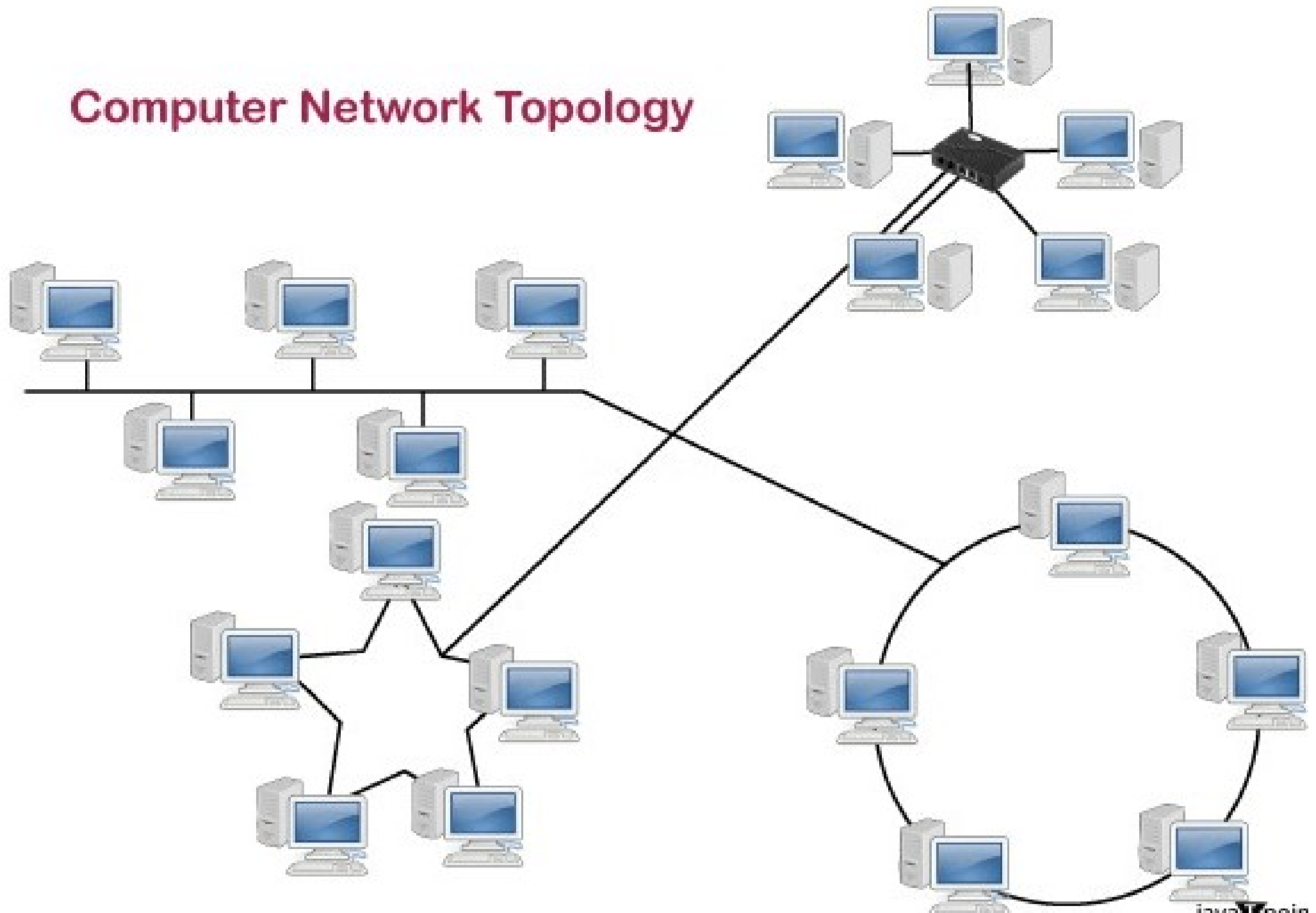
# What is Network Topology?

- **Topology** defines the structure of the network of how all the components are interconnected to each other.
- There are two types of topology:
  - Physical Topology
  - Logical Topology

# Types of Network Topology

- **Physical topology** is the geometric representation of all the nodes in a network.
- There are six types of network topology which are
  - Bus Topology
  - Ring Topology
  - Tree Topology
  - Star Topology
  - Mesh Topology
  - Hybrid Topology

# Computer Network Topology



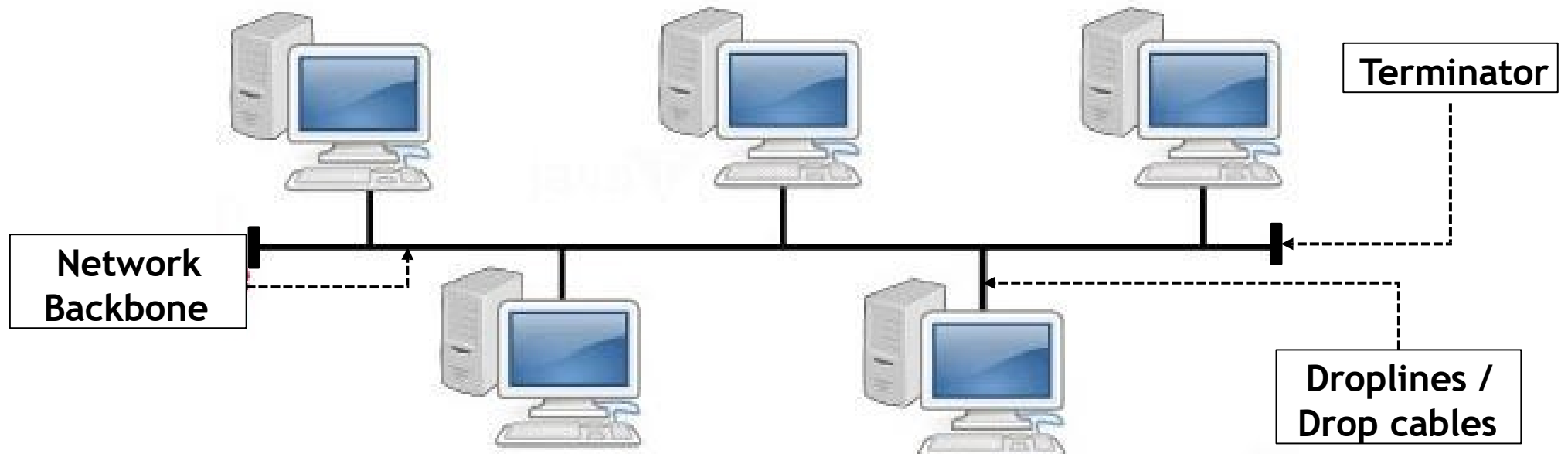
# Bus Topology

- The bus topology is designed in such a way that all the stations are connected through a single cable known as a **backbone cable**.
- Each node is either connected to the backbone cable by drop cable or directly connected to the backbone cable.
- When a node wants to send a message over the network, it puts a message over the network.
- All the stations available in the network will receive the message whether it has been addressed or not.

# Bus Topology

- The bus topology is mainly used in 802.3 (ethernet) and 802.4 standard networks.
- The configuration of a bus topology is quite simpler as compared to other topologies.
- The backbone cable is considered as a "single lane" through which the message is broadcast to all the stations.
- The most common access method of the bus topologies is **CSMA** (Carrier Sense Multiple Access).

# Bus Topology



# CSMA

- It is a media access control used to control the data flow so that data integrity is maintained, i.e., the packets do not get lost.
- There are two alternative ways of handling the problems that occur when two nodes send the messages simultaneously.
  - CSMA CD (Collision Detection)
  - CSMA CA (Collision Avoidance)



# CSMA

## CSMA CD

- CSMA CD (Collision Detection) is an access method used to detect the collision.
- Once the collision is detected, the sender will stop transmitting the data.
- Therefore, it **works** on "recovery after the collision".

## CSMA CA

- CSMA CA (Collision Avoidance) is an access method used to avoid the collision by checking whether the transmission media is busy or not.
- If busy, then the sender waits until the media becomes idle.
- This technique effectively reduces the possibility of the collision.
- It does **not work** on "recovery after the collision".

# Advantages of Bus Topology

## Low-cost Cable

- In bus topology, nodes are directly connected to the cable without passing through a hub.
- Therefore, the initial cost of installation is low.

## Moderate Data Speeds

- Coaxial or twisted pair cables are mainly used in bus-based networks that support up to 10 Mbps.

## Familiar Technology

- Bus topology is a familiar technology as the installation and troubleshooting techniques are well known, and hardware components are easily available.

## Limited Failure

- A failure in one node will not have any effect on other nodes.

# Disadvantages of Bus Topology

Extensive  
cabling:

- A bus topology is quite simpler, but still it requires a lot of cabling.

Difficult  
troubleshooting:

- It requires specialized test equipment to determine the cable faults. If any fault occurs in the cable, then it would disrupt the communication for all the nodes.

Signal  
interference:

- If two nodes send the messages simultaneously, then the signals of both the nodes collide with each other.

Reconfiguration  
difficult:

- Adding new devices to the network would slow down the network.

Attenuation:

- Attenuation is a loss of signal leads to communication issues. Repeaters are used to

# Ring Topology

- Ring topology is like a bus topology, but with connected ends.
- The node that receives the message from the previous computer retransmit to the next node.
- The data flows in one direction, i.e., it is unidirectional.
- The data flows in a single loop continuously known as an **endless loop**.
- It has no terminated ends, i.e., each node is connected to other node and having no termination point.
- The data in a ring topology flow in a clockwise direction.
- The most common access method of the ring topology is token passing.
  - **Token passing:** It is a network access method in which token is passed from one node to another node.
  - **Token:** It is a frame that circulates around the network.

# Ring Topology



# Working of Token Passing

- A token moves around the network, and it is passed from computer to computer until it reaches the destination.
- The station modifies the token by putting the address along with the data.
- The data is passed from one device to another device until the destination address matches the token received by the destination device, then it sends acknowledgment to the sender.
- In a ring topology, a token is used as a **carrier**.
- There are two types of token release techniques:
  - **Early token release** releases the token just after transmitting the data and
  - **Delayed token release** releases the token after the acknowledgment is received from the receiver.

# Advantages of Ring Topology

- Faulty devices can be removed from the network without bringing the network down.

## Network Management

- Many hardware and software tools for network operation and monitoring are available.

## Product availability

- Twisted pair cabling is inexpensive and easily available. Therefore, the installation cost is very low.

## Cost

- It is a more reliable network because the communication system is not dependent on the single host computer.

## Reliable

# Disadvantages of Ring Topology

## Difficult Troubleshooting

It requires specialized test equipment to determine the cable faults. If any fault occurs in the cable, then it would disrupt the communication for all the nodes.

## Failure

The breakdown in one station leads to the failure of the overall network.

## Reconfiguration Difficult

Adding new devices to the network would slow down the network.

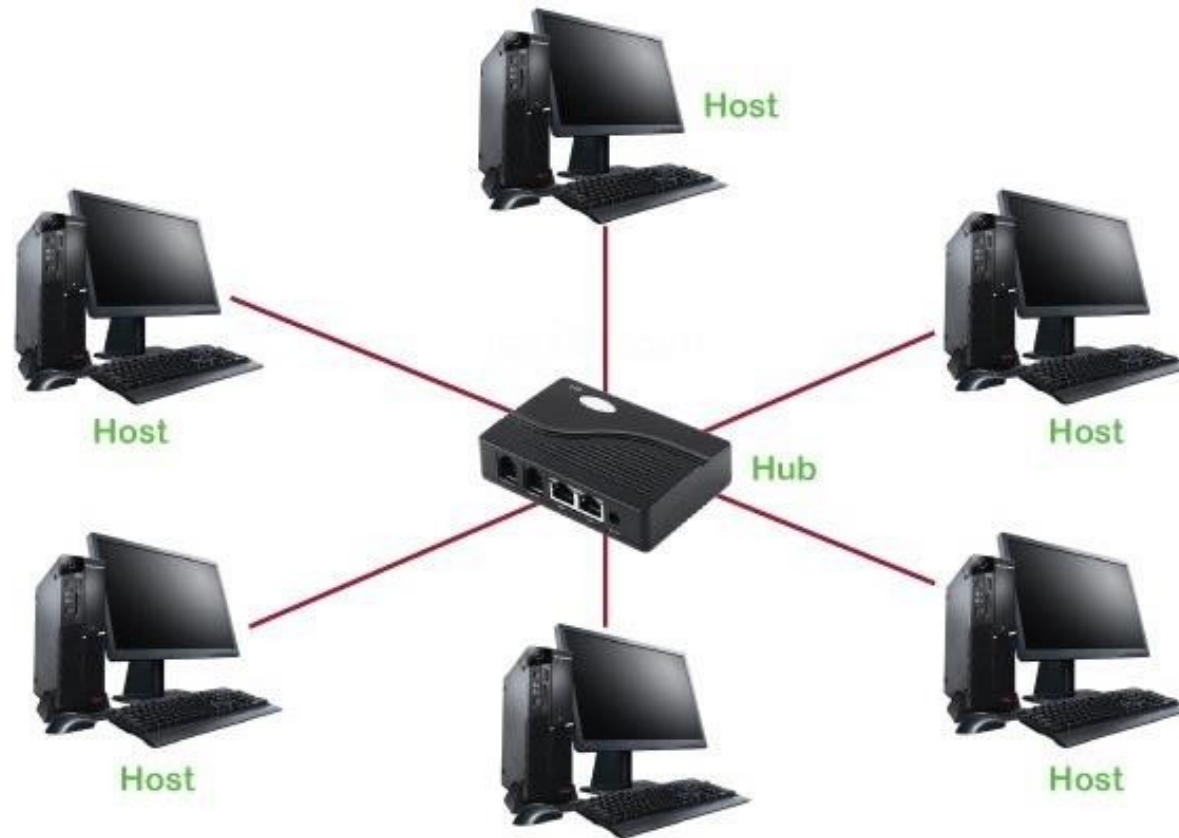
## Delay

Communication delay is directly proportional to the number of nodes. Adding new devices increases the communication delay.



# Star Topology

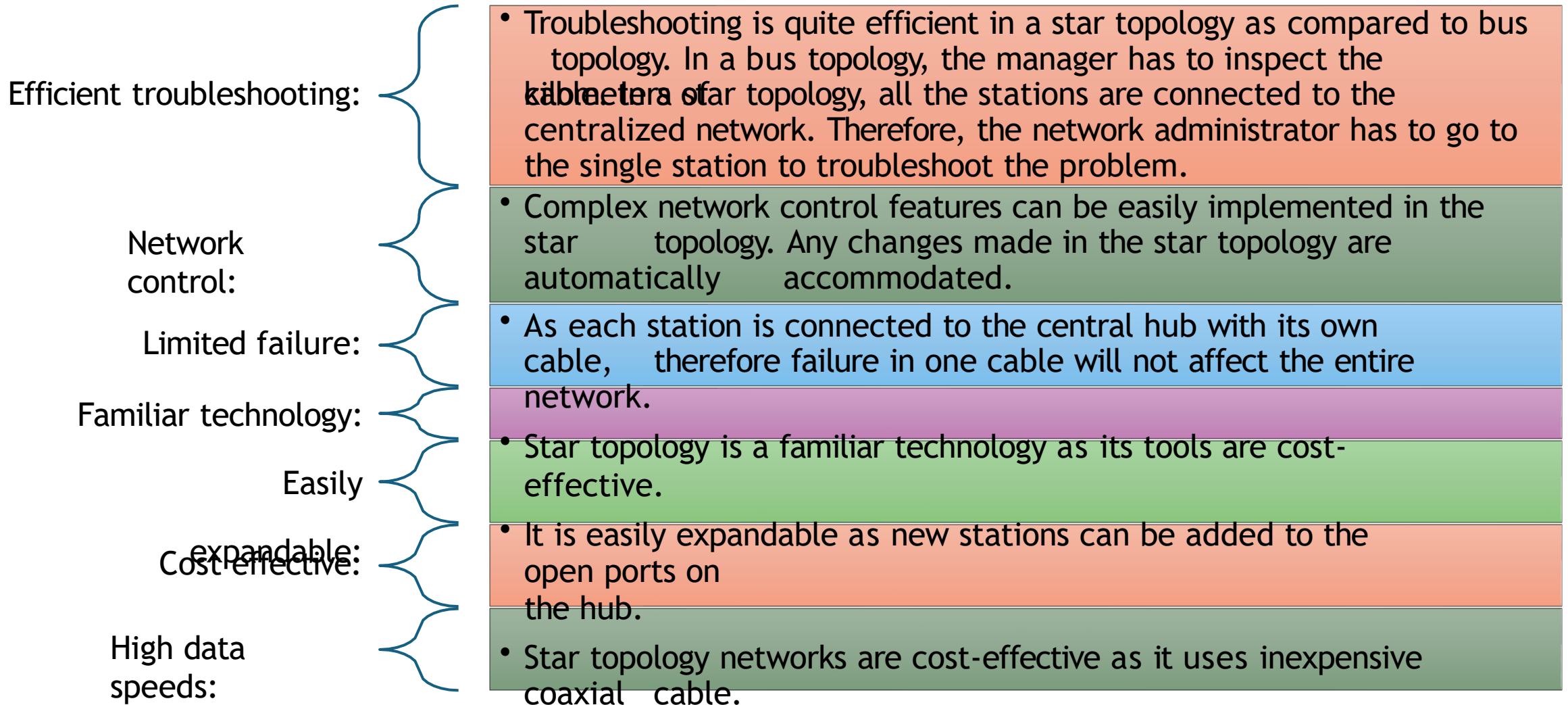
- All the hosts in Star topology are connected to a central device, known as hub device, using a point-to-point connection.



# Star Topology

- Star topology is an arrangement of the network in which every node is connected to the central hub, switch or a central computer.
- The central computer is known as a server, and the peripheral devices attached to the server are known as **clients**.
- Coaxial cable or RJ-45 cables are used to connect the computers.
- Hubs or Switches are mainly used as connection devices in a physical star topology.
- Star topology is the most popular topology in network implementation.

# Advantages of Star Topology

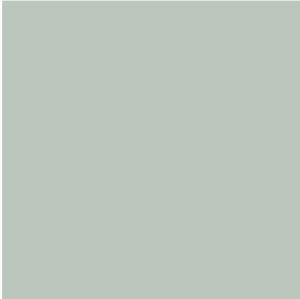


# Disadvantages of Star Topology



## A Central point of

- If the central hub or switch goes down, then all the connected nodes will not be able to communicate with each other.



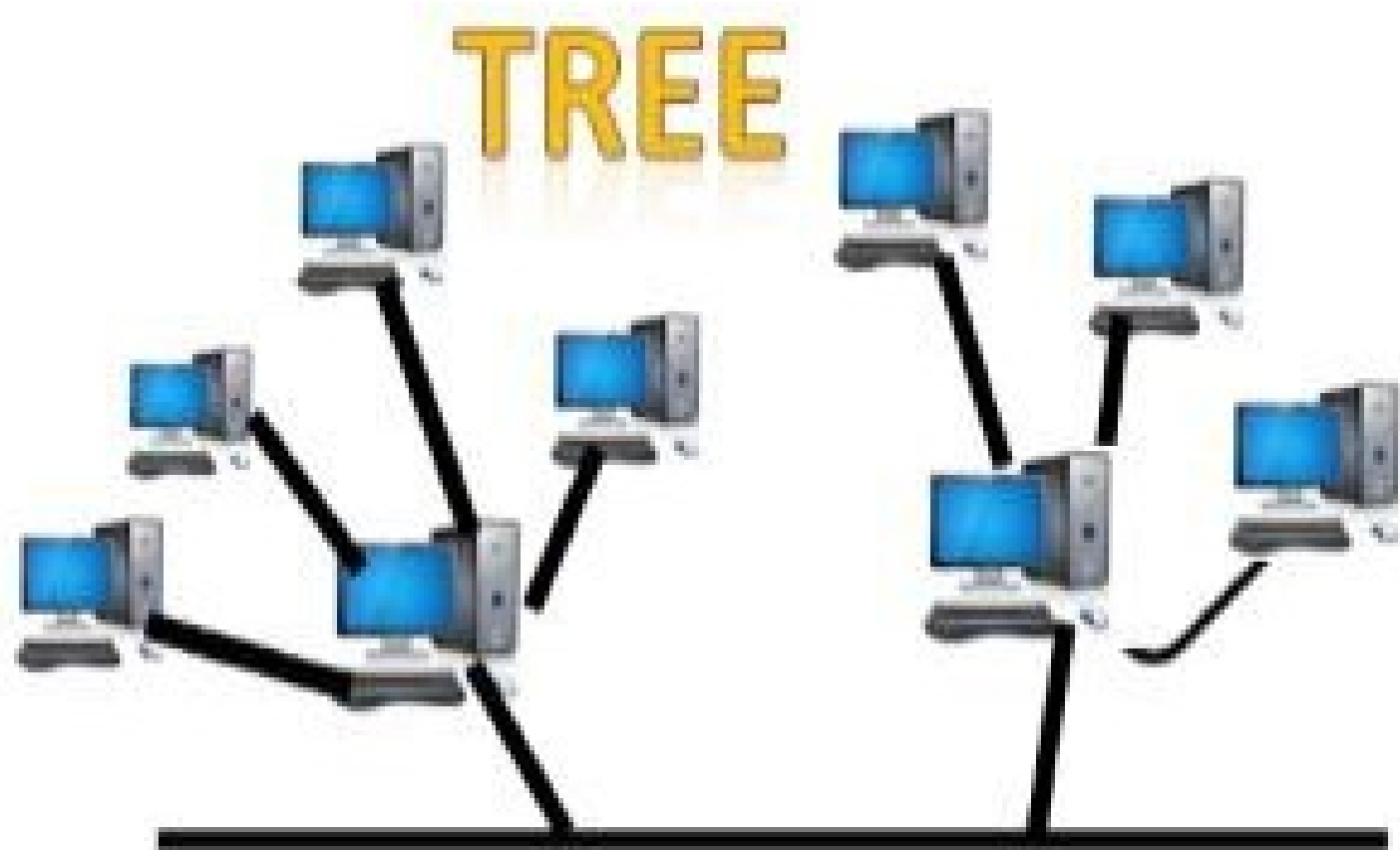
## Cable

- Sometimes cable routing becomes difficult when a significant amount of routing is required.

# Tree Topology

- Tree topology combines the characteristics of bus topology and star topology.
- A tree topology is a type of structure in which all the computers relate to each other in hierarchical fashion.
- The top-most node in tree topology is known as a **root node**, and all other nodes are the descendants of the root node.
- There is only one path exists between two nodes for the data transmission.  
Thus, it forms a *parent-child hierarchy*.

# Tree Topology



# Advantages of Tree Topology

## Support for broadband transmission

Tree topology is mainly used to provide broadband transmission, signals are sent over long distances without being

ie

attenuated.

## Easily expandable

We can add the new device to the existing network. Therefore, we can say

## Easily manageable

that tree topology is easily expandable.

In tree topology, the whole network is divided into segments known as star

## Error detection

networks which can be easily managed and maintained.

## Limited failure

Error detection and error correction are very easy in a tree topology.

## Point-to-point wiring

The breakdown in one station does not affect the entire network.

It has point-to-point wiring for individual segments.

# Disadvantages of Tree Topology

- If any fault occurs in the node, then it becomes difficult to troubleshoot the problem.

**Difficult  
Troubleshooting**

- Devices required for broadband transmission are very costly.

**High Cost**

- A tree topology mainly relies on main bus cable and failure in main bus cable will damage the overall network.

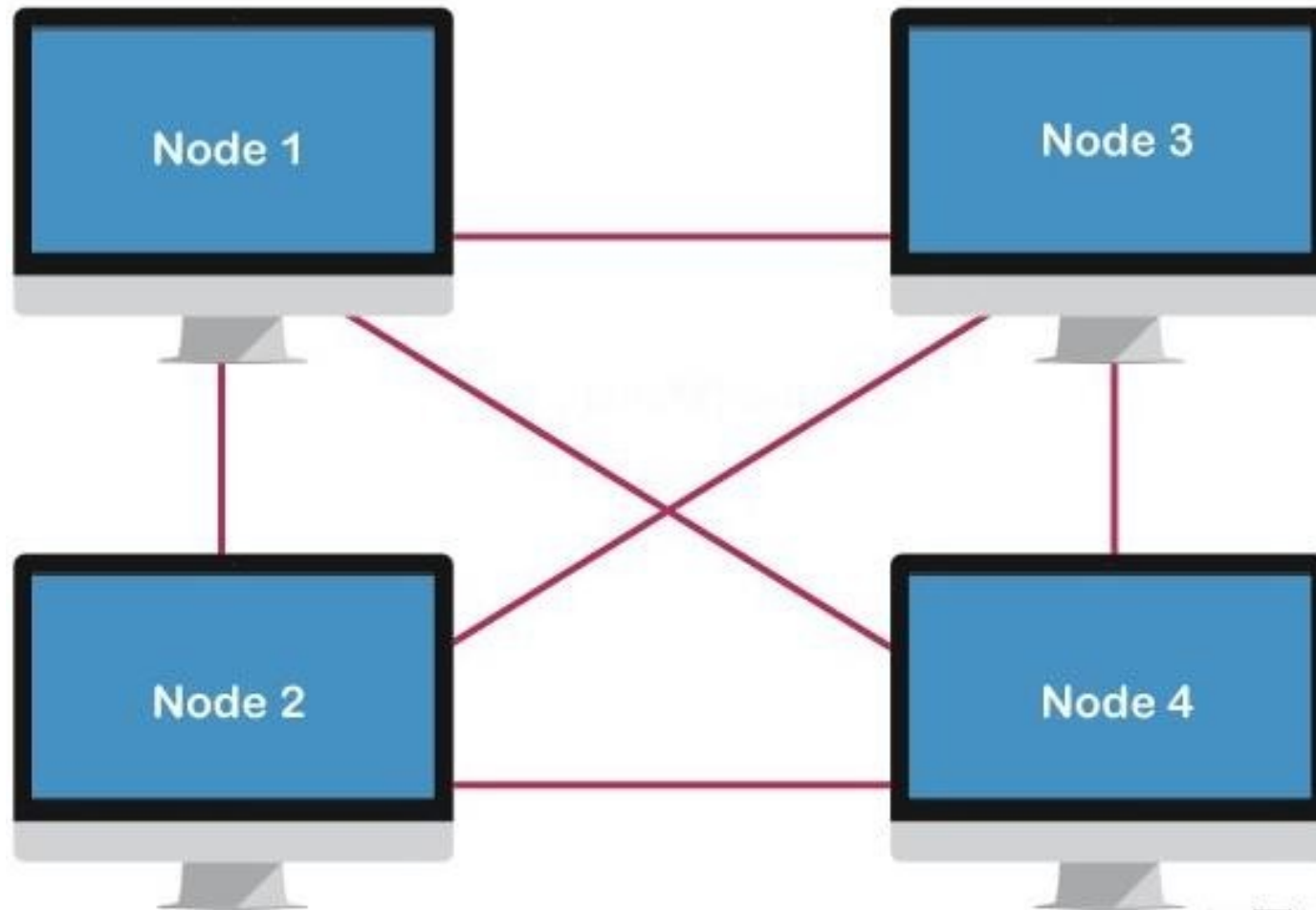
**Failure**

- If new devices are added, then it becomes difficult to reconfigure.

**Reconfiguration  
Difficult**



# Mesh Topology



# Mesh Topology

- Mesh technology is an arrangement of the network in which computers are interconnected with each other through various redundant connections.
- There are multiple paths from one computer to another computer.
- It does not contain the switch, hub or any central computer which acts as a central point of communication.
- The Internet is an example of the mesh topology.
- Mesh topology is mainly used for WAN implementations where communication failures are a critical concern.
- Mesh topology is mainly used for wireless networks.
- Mesh topology can be formed by using the formula:
  - Number of cables =  $(n*(n-1))/2$
  - Number of ports =  $n*(n-1)$
  - Where n is the number of nodes that represents the network.

# Categories of Mesh Topology

## Full Mesh Topology

- In a full mesh topology, each computer is connected to all the computers available in the network.

## Partial Mesh Topology

- In a partial mesh topology, not all but certain computers are connected to those computers with which they communicate frequently.

# Advantages of Mesh Topology

## **Reliable:**

- The mesh topology networks are very reliable as if any link breakdown will not affect the communication between connected computers.

## **Fast Communication:**

- Communication is very fast between the nodes.

## **Easier Reconfiguration:**

- Adding new devices would not disrupt the communication between other devices.

# Disadvantages of Mesh Topology

## Cost

- A mesh topology contains a large number of connected devices such as a router and more transmission media than other topologies.

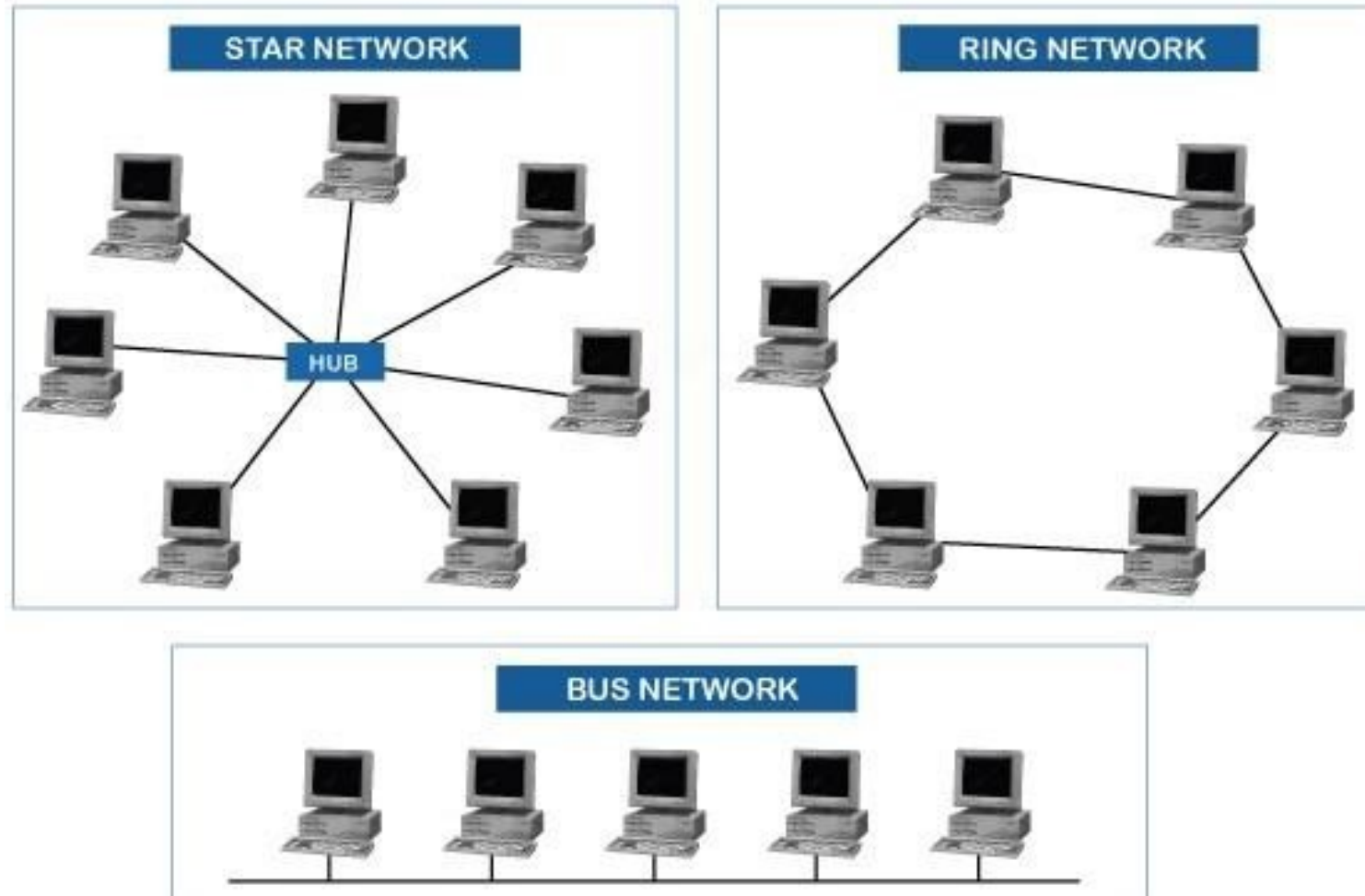
## Management

- Mesh topology networks are very large and very difficult to maintain and manage. If the network is not monitored carefully, then the communication link failure goes undetected.

## Efficiency

- In this topology, redundant connections are high that reduces the efficiency of the network.

# Hybrid Topology



# Hybrid Topology

- The combination of various different topologies is known as **Hybrid topology**.
- A Hybrid topology is a connection between different links and nodes to transfer the data.
- When two or more different topologies are combined together is termed as Hybrid topology and if similar topologies are connected with each other will not result in Hybrid topology.
  - For example, if there exist a ring topology in one branch of HBL bank and bus topology in another branch of HBL bank, connecting these two topologies will result in Hybrid topology.

# Advantages of Hybrid Topology

## Reliable

- If a fault occurs in any part of the network will not affect the functioning of the rest of the network.

## Scalable

- Size of the network can be easily expanded by adding new devices without affecting the functionality of the existing network.

## Flexible

- This topology is very flexible as it can be designed according to the requirements of the organization.

## Effective

- Hybrid topology is very effective as it can be designed in such a way that the strength of the network is maximized and weakness of the network is minimized.



# Disadvantages of Hybrid Topology

## Complex Design

- The major drawback of the Hybrid topology is the design of the Hybrid network. It is very difficult to design the architecture of the Hybrid network.

## Costly Hub

- The Hubs used in the Hybrid topology are very expensive as these hubs are different from usual Hubs used in other topologies.

## Costly Infrastructure

- The infrastructure cost is very high as a hybrid network requires a lot of cabling, network devices, etc.